

ECON 320 – Introduction to Mathematical Economics

Quiz 1 | Fall 2026

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Part A: Multiple Choice — Answer Key

Q1. Correct Answer: B

B) $\{1, 3, 4, 5, 7, 8\}$

Explanation. First compute $A \cap B = \{1, 2, 5, 6\} \cap \{2, 3, 6, 7\} = \{2, 6\}$. Then $(A \cap B)^c = U \setminus \{2, 6\} = \{1, 3, 4, 5, 7, 8\}$. Option A is just $A \cap B$ itself. Option C is $A \cup B$. Option D is neither.

Q2. Correct Answer: C

C) 32

Explanation. The power set of a set with n elements has 2^n elements. Here $n = 5$, so $|\mathcal{P}(A)| = 2^5 = 32$. Option A ($10 = \binom{5}{2}$) counts only 2-element subsets. Option B ($25 = 5^2$) confuses exponent and base. Option D ($64 = 2^6$) uses the wrong exponent.

Q3. Correct Answer: B

B) $f(x) = e^x$

Explanation. $f(x) = e^x$ is strictly increasing on $\mathbb{R}$, so it is injective. Its range is $(0, \infty) \neq \mathbb{R}$, so it is not surjective onto $\mathbb{R}$.

- **A)** x^2 : not injective since $f(-1) = f(1) = 1$.
- **C)** x^3 : bijective (both injective and surjective onto $\mathbb{R}$).
- **D)** $\sin x$: neither injective ($\sin 0 = \sin \pi$) nor surjective onto $\mathbb{R}$ (range = $[-1, 1]$).

Q4. Correct Answer: A

A) $f^{-1}(x) = \frac{x+3}{x-2}$

Explanation. Set $y = \frac{2x+3}{x-1}$ and solve for x :

$$y(x-1) = 2x+3 \Rightarrow xy - y = 2x+3 \Rightarrow xy - 2x = y+3 \Rightarrow x(y-2) = y+3 \Rightarrow x = \frac{y+3}{y-2}.$$

So $f^{-1}(x) = \frac{x+3}{x-2}$, defined for $x \neq 2$. The other options result from algebraic errors in rearranging the equation.

Q5. Correct Answer: B

B) $\sqrt{10}$

Explanation. Apply f first, then g :

$$(g \circ f)(3) = g(f(3)) = g(3^2 - 1) = g(8) = \sqrt{8+2} = \sqrt{10}.$$

Option A ($2\sqrt{2}$) would arise from computing $g(f(\sqrt{3}))$. Options C and D result from order-of-composition errors.

Part B: Numerical Problems — Full Solution

Q6. Set Operations. (5 marks)

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}, \quad A = \{1, 3, 5, 7, 9\}, \quad B = \{1, 2, 3, 4, 5\}, \quad C = \{3, 5, 6, 7, 8\}.$$

Solution

(a) $A \cap B$: elements in both A and B :

$$A \cap B = \{1, 3, 5\}.$$

$A \setminus B$: elements in A but not in B :

$$A \setminus B = \{7, 9\}.$$

(b) First, $A \cup B = \{1, 2, 3, 4, 5, 7, 9\}$.

$$(A \cup B)^c = U \setminus \{1, 2, 3, 4, 5, 7, 9\} = \{6, 8, 10\}.$$

(c) Compute A^c and B^c separately, then intersect:

$$A^c = \{2, 4, 6, 8, 10\}, \quad B^c = \{6, 7, 8, 9, 10\}.$$

$$A^c \cap B^c = \{6, 8, 10\}.$$

Since $(A \cup B)^c = \{6, 8, 10\} = A^c \cap B^c$, De Morgan's Law is verified. ✓

(d) From (a), $A \cap B = \{1, 3, 5\}$. Now intersect with $C = \{3, 5, 6, 7, 8\}$:

$$A \cap B \cap C = \{3, 5\}.$$

Since $|A \cap B \cap C| = 2$:

$$|\mathcal{P}(A \cap B \cap C)| = 2^2 = 4.$$

Q7. Inverse Functions. (5 marks)

$$f(x) = \frac{2x+3}{x-1}, \quad x \neq 1.$$

Solution**(a) Domain and Range.**

Domain: f is undefined at $x = 1$, so $\text{dom}(f) = \mathbb{R} \setminus \{1\}$.

Range: Set $y = \frac{2x+3}{x-1}$ and solve for x : $y(x-1) = 2x+3 \Rightarrow x(y-2) = y+3 \Rightarrow x = \frac{y+3}{y-2}$.

This requires $y \neq 2$, so $\text{ran}(f) = \mathbb{R} \setminus \{2\}$.

(b) Inverse Function.

From part (a):

$$f^{-1}(x) = \frac{x+3}{x-2}, \quad \text{dom}(f^{-1}) = \mathbb{R} \setminus \{2\}.$$

(c) Verification: $f(f^{-1}(x)) = x$.

$$f\left(\frac{x+3}{x-2}\right) = \frac{2 \cdot \frac{x+3}{x-2} + 3}{\frac{x+3}{x-2} - 1} = \frac{\frac{2(x+3) + 3(x-2)}{x-2}}{\frac{(x+3) - (x-2)}{x-2}} = \frac{2x+6+3x-6}{x+3-x+2} = \frac{5x}{5} = x. \quad \checkmark$$

Q8. Profit Function Analysis. (5 marks)

$$\pi(x) = -2x^2 + 12x - 10, \quad x \geq 0.$$

Solution

(a) Break-even quantities: $\pi(x) = 0$.

$$-2x^2 + 12x - 10 = 0 \Rightarrow x^2 - 6x + 5 = 0 \Rightarrow (x - 1)(x - 5) = 0.$$

Break-even outputs: $\mathbf{x = 1}$ and $\mathbf{x = 5}$.

(b) Profit-maximising quantity and maximum profit.

$$\pi'(x) = -4x + 12 = 0 \Rightarrow x^* = 3.$$

$$\pi^* = \pi(3) = -2(9) + 12(3) - 10 = -18 + 36 - 10 = \mathbf{8}.$$

(c) Concavity.

$$\pi''(x) = -4 < 0 \quad \text{for all } x \geq 0.$$

Since $\pi''(x) < 0$ everywhere, π is **strictly concave**. This confirms $x^* = 3$ is a global maximum and reflects diminishing marginal profit.

(d) Range of π on $[0, \infty)$.

Complete the square: $\pi(x) = -2(x - 3)^2 + 8$.

At $x = 0$: $\pi(0) = -10$. Maximum at $x^* = 3$: $\pi(3) = 8$. As $x \rightarrow \infty$: $\pi(x) \rightarrow -\infty$.

$$\text{Range of } \pi = (-\infty, 8].$$